

Google

You are a Product Analyst on the Google Pay security team focused on improving the reliability of digital payments....

Question 1: For January 2024, what are the total counts of successful and failed transactions in each merchant category? This analysis will help the Google Pay security team identify potential friction points in payment processing.

```
SELECT
    merchant_category,
    SUM(CASE WHEN transaction_status = 'successful' THEN 1 ELSE 0 END) AS successful_transactions,
    SUM(CASE WHEN transaction_status = 'failed' THEN 1 ELSE 0 END) AS failed_transactions
FROM fct_transactions
WHERE transaction_date >= '2024-01-01'
    AND transaction_date < '2024-02-01'
GROUP BY merchant_category
ORDER BY merchant_category;
```



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Question 2: For the first quarter of 2024, which merchant categories recorded a transaction success rate below 90%? This insight will guide our prioritization of security enhancements to improve payment reliability.

```
SELECT
    merchant_category,
    COUNT(CASE WHEN transaction_status = 'successful' THEN 1 END) *
    100.0
        / COUNT(*) AS success_rate
FROM fct_transactions
WHERE transaction_date >= '2024-01-01'
    AND transaction_date < '2024-04-01'
GROUP BY merchant_category
HAVING COUNT(CASE WHEN transaction_status = 'successful' THEN 1 END)
    * 100.0
        / COUNT(*) < 90
ORDER BY success_rate;
```



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Question 3: From January 1st to March 31st, 2024, can you generate a list of merchant categories with their concatenated counts for successful and failed transactions? Then, rank the categories by total transaction volume. This ranking will support our assessment of areas where mixed transaction outcomes may affect user experience.

```
SELECT
  merchant_category,
  CONCAT(
    SUM(CASE WHEN transaction_status = 'successful' THEN 1 ELSE
0 END),
    ' successful, ',
    SUM(CASE WHEN transaction_status = 'failed' THEN 1 ELSE 0 EN
D),
    ' failed'
  ) AS transaction_counts,
  COUNT(*) AS total_transaction_volume,
  RANK() OVER (ORDER BY COUNT(*) DESC) AS transaction_volume_rank
FROM fct_transactions
WHERE transaction_date >= '2024-01-01'
  AND transaction_date < '2024-04-01'
GROUP BY merchant_category
ORDER BY transaction_volume_rank;
```

